

Software Requirements Specification

ProvStock – Prov&Tec Management: Comprehensive Sales and Inventory System

V1.1

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of ProvStock, the Comprehensive Sales and Inventory System for Prov&Tec.

This document describes the system's expected functionalities, user interactions, and operational constraints. It serves as a reference for the development team and stakeholders throughout the software development process, including system design, implementation, and testing.

The SRS aims to establish a clear understanding of what the system must accomplish and to make sure that its development addresses the sales and inventory management needs identified in the project.

This document presents an initial proposal of software requirements. The proposed scope and requirements are subject to review, validation, and changes following feedback from the client, Ximena Añamise.

1.2 Scope

ProvStock is a software solution designed to organize and manage the sales and inventory operations of Prov&Tec, a company that sells electrical materials.

The system will centralize information about customers, products, suppliers, sales, and purchases. It will allow users to register and update records, process sales, record purchases made to suppliers, and keep the stock of each product updated every time a sale or a purchase is registered.

Additionally, the system will provide access to the history of registered sales, generate alerts when a product reaches or falls below its minimum stock level, and produce reports about sales, inventory, and purchases.

The primary objective is to replace the manual, paper-based management currently used by the company, reduce the time customers wait to be served, avoid unnecessary purchases of materials already in stock, and support the owner's control and decision-making.

1.3 Definitions and Acronyms

The following acronyms are used throughout this document. Project terminology is defined in Appendix A.

Term	Definition
SRS	Software Requirements Specification. A document that defines the requirements of a software system.
FR	Functional Requirement. A requirement that describes a function the system must perform.
NFR	Non-Functional Requirement. A requirement that describes a quality attribute or operational constraint of the system.

RF	Functional Requirement (Requisito Funcional). Name used for the features of the project Product Backlog, RF01 to RF09.

1.4 References

The following project documents provide the foundation for this Software Requirements Specification:

Byte Works. Workshop 07 – Project Definition: Prov&Tec Management: Comprehensive Sales and Inventory System. University of the Armed Forces – ESPE, 2026.

Byte Works. Product Backlog: ProvStock, requirements RF01–RF09. University of the Armed Forces – ESPE, 2026.

2. Overall Description

2.1 Product Perspective

ProvStock is a software solution designed to centralize and organize the sales and inventory operations of Prov&Tec, replacing its current manual management.

Currently, the company manages its sales and inventory with paper documents, such as sales slips and order forms, and basic spreadsheets. This makes it hard to know the real stock of each product, so the recording of incoming and outgoing goods is slow and inconsistent. As a result, staff lose time looking for products, materials that are already in stock are ordered again, and customers sometimes wait between 10 and 20 minutes to be served. The proposed system aims to solve these limitations by providing a single environment for managing this information.

The system will integrate user access control, customer, product, and supplier management, sales and purchase registration, automatic inventory updates, a sales history, stock alerts, and reports.

By organizing these operations within a single system, the software is expected to improve information accessibility, customer service speed, inventory control, and the efficiency of the business.

2.2 Product Functions

The main functions of ProvStock are described below:

Access Management: Define user types and control access to the system through login and password management.

Customer Management: Register, update, and view customer information.

Product Management: Register, update, and view the electrical products sold by the company.

Sales Management: Register sales, record the customer of each sale, and select the products included in it.

Inventory Management: Control product stock and update it automatically when a sale or a purchase is registered.

Supplier and Purchase Management: Register and manage suppliers, register purchases, and record the products included in each one.

Sales History: View the complete history of registered sales, with the date and the salesperson responsible.

Alerts and Notifications: Monitor stock levels and identify products that reach or fall below the minimum stock.

Reports: Generate reports about sales, current inventory, and purchases.

These functions correspond to the nine features (RF01 to RF09) established in the Product Backlog of the project.

2.3 User Classes and Characteristics

The system is intended to support three main user roles:

Salesperson

The salesperson attends the customers at the store. This role uses the system to register sales, select the products requested by the customer, and consult or register customer information.

Warehouse Clerk

The warehouse clerk is in charge of the warehouse. This role controls the stock of each product, registers the products received from suppliers, and receives the alerts about products with low stock.

Owner

The owner is responsible for the business as a whole. This role manages supplier information, registers purchases, reviews the sales history, and consults the reports and alerts to make decisions. The owner is also the client who validates this document.

In addition, the stock monitoring that generates the alerts is performed automatically by the system, without a user starting it. The use case diagram that shows how each role interacts with the system is included in Appendix D.

2.4 Operating Environment

ProvStock is intended to operate in a computer environment in which the sales, purchase, customer, product, and supplier information can be registered, stored, retrieved, and managed.

The system will require access to a data storage mechanism to keep the records of users, customers, products, suppliers, sales, purchases, and inventory.

Its user interface should allow the salespeople, the warehouse clerk, and the owner to carry out their tasks quickly and to consult the registered information.

The system is expected to be developed with object-oriented programming, using Java as in the course. The specific operating system, database management system, deployment platform, and hardware requirements have not yet been established and must be defined during the technical design phase.

2.5 Design and Implementation Considerations

The system design should maintain consistency among customers, products, sales, purchases, suppliers, and inventory records.

Each sale and each purchase must be connected to the products it includes, and the stock of those products must be updated automatically when the operation is registered.

The software should also keep the date and the salesperson of every sale so that the sales history can be consulted at any time.

The implementation must support the nine features established in the Product Backlog, respecting their defined priorities.

2.6 Assumptions and Dependencies

The proposed system assumptions and operational dependencies are specified in Section 7. They will be reviewed during stakeholder validation.

3. Functional Requirements

This section describes the main functional requirements of ProvStock. The requirements are based on the project's Product Backlog and define the essential functions proposed for the system.

3.1 FR-01: Access Management

Description: The system shall allow controlled access to its functions according to user types.

Functional Requirements:

- FR-01.1: The system shall allow defining user types (Salesperson, Warehouse Clerk, and Owner) according to their access and responsibilities.
- FR-01.2: The system shall require users to log in with a username and password.
- FR-01.3: The system shall allow users to manage their password to protect their accounts.

Priority: High

Product Backlog Reference: RF01

Acceptance Criteria: Users can log in with valid credentials, invalid attempts are rejected, and each user can only access the functions allowed for their role.

3.2 FR-02: Customer Management

Description: The system shall allow users to manage customer information.

Functional Requirements:

- FR-02.1: The system shall allow users to register new customers by entering the required customer information.
- FR-02.2: The system shall allow users to edit and update existing customer data.
- FR-02.3: The system shall allow users to view the registered customer information.

Priority: High

Product Backlog Reference: RF02

Acceptance Criteria: Users can register, update, and view customer records.

3.3 FR-03: Product Management

Description: The system shall allow users to manage the products sold by the company.

Functional Requirements:

- FR-03.1: The system shall allow users to register products by entering the required information through a product form.
- FR-03.2: The system shall allow users to edit and update product information.
- FR-03.3: The system shall allow users to view the products registered in the system.

Priority: High

Product Backlog Reference: RF03

Acceptance Criteria: Users can register, update, and view product records.

3.4 FR-04: Sales Management

Description: The system shall allow salespeople to register sales.

Functional Requirements:

- FR-04.1: The system shall allow salespeople to register new sales in the system.
- FR-04.2: The system shall record the customer associated with each sale.
- FR-04.3: The system shall allow users to select the products, and their quantities, included in the sale.
- FR-04.4: The system shall calculate the total amount of each sale from the selected products.

Priority: High

Product Backlog Reference: RF04

Acceptance Criteria: A salesperson can register a sale with its customer, its selected products, and the total amount.

3.5 FR-05: Inventory Management

Description: The system shall allow the control of product stock.

Functional Requirements:

- FR-05.1: The system shall allow the warehouse clerk to control and manage the stock of each product.
- FR-05.2: The system shall automatically update the product stock when a sale is registered.
- FR-05.3: The system shall automatically update the product stock when a purchase is registered.

Priority: High

Product Backlog Reference: RF05

Acceptance Criteria: The stock of a product decreases after a registered sale and increases after a registered purchase, and the warehouse clerk can consult the current stock.

3.6 FR-06: Supplier and Purchase Management

Description: The system shall allow users to manage suppliers and register purchases.

Functional Requirements:

- FR-06.1: The system shall allow the owner to register and manage supplier information.
- FR-06.2: The system shall allow users to register new purchases in the system.

- FR-06.3: The system shall record the products included in each purchase.

Priority: High

Product Backlog Reference: RF06

Acceptance Criteria: The owner can register and update suppliers, and users can register purchases with their products.

3.7 FR-07: Sales History

Description: The system shall allow the owner to consult the registered sales.

Functional Requirements:

- FR-07.1: The system shall allow the owner to view the complete history of registered sales.
- FR-07.2: The system shall record the date on which each sale was made.
- FR-07.3: The system shall display the salesperson responsible for each sale.

Priority: High

Product Backlog Reference: RF07

Acceptance Criteria: The owner can view all registered sales with their date and responsible salesperson.

3.8 FR-08: Alerts and Notifications

Description: The system shall monitor stock levels and generate alerts.

Functional Requirements:

- FR-08.1: The system shall monitor the stock level of the registered products.
- FR-08.2: The system shall identify the products that reach the minimum stock level.
- FR-08.3: The system shall identify the products whose stock falls below the minimum level.
- FR-08.4: The system shall display an alert in the application for each identified product.

Priority: High

Product Backlog Reference: RF08

Acceptance Criteria: When a product reaches or falls below its minimum stock, an alert for that product is displayed to the warehouse clerk and the owner.

3.9 FR-09: Reports

Description: The system shall generate reports of the business operations.

Functional Requirements:

- FR-09.1: The system shall generate reports about the registered sales.
- FR-09.2: The system shall generate reports about the current inventory and stock levels.
- FR-09.3: The system shall generate reports about the registered purchases.

Priority: High

Product Backlog Reference: RF09

Acceptance Criteria: Users can generate the sales, inventory, and purchase reports with the information recorded in the system.

The relationship between functional requirements and Product Backlog items is provided in Appendix B. These requirements remain subject to stakeholder feedback.

4. Non-Functional Requirements

This section describes the quality requirements of ProvStock. These requirements define the expected performance, security, usability, reliability, and maintainability of the proposed software.

4.1 NFR-01: Performance

The system shall respond to basic operations, such as registering, searching, and updating records, within 3 seconds under normal operating conditions.

Acceptance Criteria: Basic operations are completed within 3 seconds during testing with sample data.

4.2 NFR-02: Security

The system shall require users to log in with a username and password before accessing any information, and shall only show each user the functions allowed for their role.

Acceptance Criteria: Users can access the system with valid credentials, invalid login attempts are rejected, and a user cannot open functions that do not belong to their role.

4.3 NFR-03: Usability

The system shall provide a simple and understandable user interface that allows salespeople to register a sale quickly and users to navigate between the main functions.

Acceptance Criteria: Users can locate and access the main system functions through clearly labeled menus and buttons.

4.4 NFR-04: Reliability

The system shall store the registered information and preserve it when the application is closed and reopened, keeping the stock of each product consistent with the registered sales and purchases.

Acceptance Criteria: Previously saved records remain available after restarting the application, and the stock shown matches the registered operations.

4.5 NFR-05: Maintainability

The system shall use an organized object-oriented structure with separate classes representing its main entities.

Acceptance Criteria: The source code contains clearly identified classes with defined responsibilities, allowing individual components to be modified without unnecessarily affecting unrelated components.

5. External Interface Requirements

This section describes the external interfaces required for ProvStock. These interfaces define how users interact with the application and how the system communicates with the hardware and software components necessary for its operation.

5.1 User Interfaces

The system shall provide a simple and organized graphical user interface that allows users to access its main functionalities.

- **UI-01:** The system shall provide a main menu to access the customer, product, sales, inventory, supplier and purchase, sales history, alerts, and report functions according to the user role.
- **UI-02:** The system shall provide forms to register and update information, such as customers, products, suppliers, sales, and purchases.
- **UI-03:** The system shall display registered information through organized lists and tables.
- **UI-04:** The system shall display clear messages to confirm completed operations, identify input errors, and show the stock alerts.

5.2 Hardware Interfaces

The system shall operate on a computer with the necessary hardware resources to execute the application.

- **HI-01:** The system shall support standard input devices, such as a keyboard and mouse.
- **HI-02:** The system shall display information through a computer monitor.
- **HI-03:** The system shall use the storage device of the computer to save the registered information.

5.3 Software Interfaces

The system shall interact with the software components necessary to manage and store sales and inventory information.

- **SI-01:** The system shall use a data storage mechanism to save and retrieve registered information.
- **SI-02:** The system shall provide access to the stored sales, purchase, and inventory information through its application interface.

5.4 Communication Interfaces

The initial proposal does not require integration with external network services, such as online sales or electronic invoicing. All the information is registered and consulted through the application, as described in FR-01 to FR-09.

6. Constraints

This section describes the main constraints that must be considered during the development of ProvStock. These constraints establish the proposed technical and operational limitations of the project.

6.1 Technical Constraints

- **CON-01:** The system shall be developed using object-oriented programming principles to organize its main functionalities.
- **CON-02:** The system shall use a simple data storage mechanism to manage users, customers, products, suppliers, sales, purchases, and inventory records.

6.2 Operational Rules

- **CON-03:** Users must provide the necessary information when registering customers, products, suppliers, sales, and purchases.
- **CON-04:** Each sale must be associated with a registered customer, a registered salesperson, and at least one registered product.
- **CON-05:** Each purchase must refer to a registered supplier and to at least one registered product.
- **CON-06:** A sale cannot include a quantity of a product greater than the stock available in the inventory.

6.3 Project Scope Constraints

- **CON-07:** The proposed scope is limited to the nine Product Backlog features; advanced extensions, such as online sales, payments, and electronic invoicing, are excluded from the initial version.

These constraints represent the initial proposed limitations of the system and may be adjusted based on the development team's capabilities and stakeholder feedback.

7. Assumptions and Dependencies

This section describes the assumptions and dependencies considered for the development and operation of ProvStock.

7.1 Assumptions

The following assumptions are considered for the proposed system:

- **AS-01:** Users are expected to provide accurate information when registering customers, products, suppliers, sales, and purchases.
- **AS-02:** Users are expected to have basic computer skills to operate the system.
- **AS-03:** Salespeople and the warehouse clerk are expected to register sales and purchases when they happen, so the stock stays up to date.
- **AS-04:** A minimum stock level is expected to be defined for each product.

7.2 Dependencies

The system's operation will depend on the following conditions:

- **DEP-01:** Sales registration will depend on the customers and products previously registered in the system.
- **DEP-02:** Inventory updates will depend on the sales and purchases registered in the system.
- **DEP-03:** Stock alerts will depend on the minimum stock level registered for each product.
- **DEP-04:** The sales history and the reports will depend on the sales, purchases, and stock information previously recorded.

- **DEP-05:** Access to each function will depend on the user types defined in FR-01.

These assumptions and dependencies represent the initial conditions considered for the proposed software and may be reviewed according to stakeholder feedback.

8. Appendices

This section provides supplementary information to support the understanding and organization of the Software Requirements Specification (SRS) for ProvStock.

Appendix A: Glossary

The following terms describe the main concepts used in the proposed system.

Term	Definition
Prov&Tec	The company that sells electrical materials and will use the system.
Customer	A person or organization that buys products from the company.
Product	An electrical material sold by the company and registered in the system.
Supplier	A person or company from which the business buys products.
Sale	A record of the products sold to a customer by a salesperson on a given date.
Purchase	A record of the products bought from a supplier.
Inventory	The stock of each product available in the warehouse.
Stock	The quantity of units of a product currently available.
Minimum Stock Level	The lowest quantity of a product that should remain in the warehouse before it has to be restocked.
Alert	A notification that a product has reached or fallen below its minimum stock level.
Sales History	A collection of all the sales registered in the system.
Report	A summary of sales, inventory, or purchase information recorded in the system.
Salesperson	A user who attends customers and registers sales.
Warehouse Clerk	A user in charge of controlling the stock in the warehouse.
Owner	The user responsible for the business, who manages suppliers and reviews history, alerts, and reports.

Appendix B: Requirements Traceability Matrix

The Requirements Traceability Matrix establishes the relationship between the functional requirements defined in this document and the features identified in the Product Backlog.

Requirement ID	Functional Requirement	Product Backlog	Priority

FR-01	Access Management	RF01	High
FR-02	Customer Management	RF02	High
FR-03	Product Management	RF03	High
FR-04	Sales Management	RF04	High
FR-05	Inventory Management	RF05	High
FR-06	Supplier and Purchase Management	RF06	High
FR-07	Sales History	RF07	High
FR-08	Alerts and Notifications	RF08	High
FR-09	Reports	RF09	High

This matrix ensures that each functional requirement corresponds to a feature established in the Product Backlog. It also facilitates future requirement reviews, modifications, and validation with stakeholders.

Appendix C: Requirement Validation

The requirements presented in this document constitute Version 1.0 of the initial proposal for ProvStock.

The proposed requirements will be reviewed with the client, Ximena Añamise, to determine whether the system addresses the sales and inventory management needs of Prov&Tec.

Any changes requested will be evaluated and incorporated into future versions of the SRS when appropriate.

The final requirements will be established after stakeholder review and validation.

Appendix D: Use Case Diagram

The following diagram shows the actors of ProvStock and the use cases they can perform. The number in parentheses indicates the Product Backlog requirement from which each use case comes. The Clock actor represents the automatic stock monitoring of FR-08.

